Learn recovery that still satisfies the chemical task

A proposed mechanism within an agentic Real–Sim–Real engineering system

ENGINEERING CONTEXT

Ingest & preflight → Build / calibrate twin → Compose or train → Deploy & diagnose

failures guide the next engineering experiment

1 Event history & contract

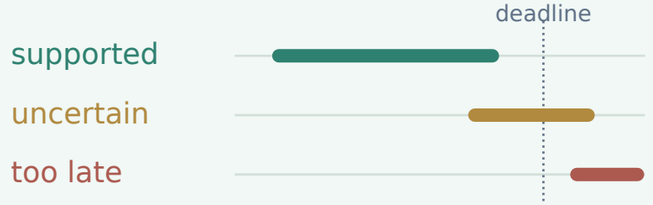
Delivered images + action feedback
Reference event + allowed window

Public estimate: phase, delay, remaining time and uncertainty

Past violations stay in the attempt record

2 Which recovery can still work?

Check duration, dependencies and support



Forecasts guide action; execution needs evidence

3 One history-conditioned policy

Execute / Observe / Recover
or safely terminate the attempt

Shared option decisions
+ short robot action chunks

No online expert in retained-learning tests

real process time

Physical execution + fixed adjudication

Outcome AND event timing
TRUE / FALSE / UNKNOWN · all attempts

Trajectory ledger D

Student states · valid recovery · failures
Consumer-specific labels and admission

Engineering artifact ledger K

Validated interfaces, perception, twin, skills
Applicability and versioned test evidence

student-state labels → policy update

Attribute gains separately: engineering reuse K · context adaptation M · actor absorption θ

Concept only. Interval bars are schematic. No new training, chemical rollout or physical result.